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Model Validation of Articulated Heavy-Duty Vehicle via IPG for Tracking Controller Design

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Abstract

Articulated heavy-duty vehicles (HDVs) exhibit complex dynamic behavior during operation, leading to several control challenges. A critical aspect that plays an important role in the vehicle's performance is the vehicle model used in the development of autonomous functions. A model that adequately represents the response of the vehicle is essential to obtain the desired dynamic simulations and implement control and planning algorithms. This paper focuses on the hybrid-Kinematic (hK) vehicle model for modelling autonomous HDVs, and the validation of its behaviour via IPG TruckMaker®. A comparison between the hK and a dynamic model was conducted to analyze and evaluate its precision and performance in reversing. The results have shown that the hK model effectively mimics the behavior of the articulated truck, thus facilitating to solve the tracking control problem.

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1. Introduction

Trajectory control of the autonomous heavy-duty vehicle (aHDV) has gained the interest of control engineers due to their challenging and complex behavior. This is due to the fact that aHDV has a complex nature because of: i) inferior maneuverability at low speeds ii) a wide range of loading conditions (S. N. Singh, 1986), iii) limited stability regions, iv) jack-knifing, v) snaking at high speeds, vi) phase lag, viii) rearward amplification (Wang, L. (n.d.)). Furthermore, difficulties regarding the control of an aHDV in forward motion for high-speed manoeuvres are discussed thoroughly in (B. Jacobson, 2015; S. Kharrazi, 2012; X. Kang and W. Deng, 2007). In the literature, various control

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techniques are applied to solve these challenging problems including both classical non-linear control techniques and intelligent controllers (Wang, L. and Huang, B., 2013).

Despite the extensive literature on various control techniques applied to aHDVs, one of the main problems with heavy vehicles is still to represent the effects of vehicle dynamics as realistically as possible using the simplest approach (X. Liu, et al. 2019). There are numerous techniques to model the behaviour of a vehicle and are principally categorized based on linearity and the response they can capture (Table 1). All models described can be defined as either four wheels or the left and right wheels lumped together into a single wheel, known as the bicycle model.

Table 1. Vehicle modelling techniques

Modelling	Linearity	States	Significance
Geometric	Linear	Vehicle dimension and position	It correlates vehicle dimension, the radius of turn and radius of road curvature/ vehicle position w.r.t. trajectory and its subsequent error (P. Ridley and P. Corke, 2003)
Kinematic	(Non-) Linear	Vehicle position, velocity, and acceleration	It describes the vehicle's lateral velocity and acceleration and yaw w.r.t. vehicle local coordinate. A geometric model generally accompanies it to test the controller performance (e.g., deviation error). (Kim and Oh, 1999)
Hybrid-Kinematic	Non-Linear	Vehicle position, velocity, and acceleration.	It describes the vehicle's lateral velocity, acceleration and yaw. It makes considerations for the finite response time of primary inputs, better capturing true vehicle response. (C. Pradalier and K. Usher, 2008)
Dynamic	Non-Linear	Vehicle position, velocity, and acceleration	It considers internal forces and momentum and better captures the response of the vehicle. It can also consider (non-) linear tires forces. The non-linear model performs better in capturing the vehicle response. (A. Mendes, 2022)

However, there is limited literature on validating the vehicle model of an aHDV. An example of model validation of an aHDV using experimental results can be found in (E Kayacan et al., 2018), in which an adaptive kinematic model was implemented with an MVP-based trajectory tracking controller. The main objective of this paper is to present and validate the hybrid-Kinematic (hK) model in conjunction with a controller in various test cases. Since a simple and accurate model is one of the main goals, the focus of the is to evaluate the hK model, which can be used to mimic the complex behavior of an actual truck trailer, especially in reverse at low speeds. The hK modelling approach was also implemented in (K. Yoshiaki, et al. 2009; Y. Kuwata, et al. 2008) for passenger vehicles.

Therefore, the focus of this paper is to evaluate the validity of the hK vehicle model for the development of automated driving functions for articulated heavy-duty vehicles. The rest of the paper is organized as follows. Section 2 presents the hK vehicle model implemented in MATLAB/Simulink along with a co-simulation model using high-fidelity IPG TruckMaker model used in the validation process (IPG TruckMaker, version 6.0.6). The trajectory controller used with both models to overcome instability is presented in Section 3. The simulation results for both models are compared and analyzed in Section 4. Section 4 summarizes the key findings and future work.

Nomenclature

x_t	Truck coordinate on the X-axle (m)
y_t	Truck coordinate on the Y-axle (m)
ψ_t	Truck yaw angle (rad)
ψ_{st}	Trailer yaw angle (rad)
γ	Hitch angle (rad)
x_{st}	Trailer coordinate on the X-axle (m)
y_{st}	Trailer coordinate on the Y-axle (m)
v_t	Truck Speed (m/s)
L_t	Truck wheelbase (m)
L_{rh}	Distance between the truck rear axle and Hitch point (m)
L_{st}	Distance between trailer middle axle and Hitch point (m)
τ_v	Longitudinal speed time constant (s)
τ_δ	Steering system time constant (s)

2. Methodology

In this section, both the non-linear hK model and IPG TruckMaker vehicle model are presented. A trajectory controller is also described, which is used to stabilize the aHDV in reverse motion. Those models will be then simulated in MATLAB/Simulink and co-simulation model to validate (Section 3).

2.1. Hybrid-kinematic vehicle model

The single-track model of an articulated vehicle is shown in Fig. 1. The assumptions made in the derivation of the model, are: i) both the truck and trailer are considered rigid bodies, ii) the tires on both sides of the truck and trailer axles are lumped together (single track model), iii) tire slip angles are small, iv) the articulation angle between truck and trailer is small, v) the longitudinal speeds of the truck and trailer are the same, vi) the longitudinal speeds are constant (zero longitudinal acceleration), vii) low lateral speed, viii) Only the motion in the x-y plane is considered (i.e. neglecting pitch and roll), and ix) coupling between longitudinal and lateral motion is neglected. Based on these assumptions the mathematical expressions used to incorporate the linear transfer functions and non-linear elements of the steering and throttle to build the hK can be written as:

$$\begin{aligned}\dot{x}_t &= v_t \cos \psi_t, & \dot{y}_t &= v_t \sin \psi_t, & \dot{\psi}_t &= \frac{v_t}{L_t} \tan \delta \\ \dot{\psi}_{st} &= \frac{v_t}{L_{st}} \sin \gamma + \left(\frac{L_{rh}}{L_{st}} \right) \left(\frac{v_t}{L_t} \cos \gamma \tan \delta \right) \\ \dot{\gamma} &= -\frac{v_t}{L_{st}} \sin \gamma + \frac{v_t}{L_t} \left(1 - \frac{L_{rh}}{L_{st}} \cos \gamma \right) \tan \delta \\ \dot{v}_t &= \frac{1}{\tau_v} (v_c - v_t), \\ \dot{\delta} &= \frac{1}{\tau_\delta} (\delta_c - \delta)\end{aligned}\quad (1)$$

where v_t and δ are the speed and the steering angle and the description of the rest of the parameters is already given in the nomenclature.

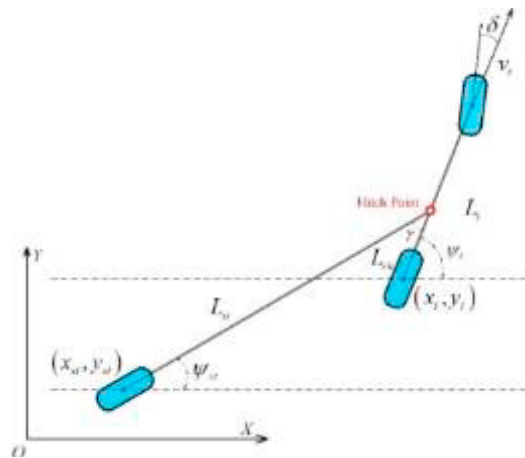


Fig. 1. Single-track bicycle model representation of an articulated vehicle

2.2. Dynamic vehicle model

A Ford Otosan truck-trailer is modelled in the IPG TruckMaker® software, as seen in Fig. 2. The vehicle model is a multi-body system which is characterized by different bodies containing all sprung masses except the engine, trim loads and additional loads, the suspension, which includes all unsprung masses and finally, the 4 wheels. External and internal forces/torques and constraints are determined by: i) Suspension Force Elements, ii) Aerodynamics, iii) Kinematics and Compliance, iv) Tire forces/torques v) User-defined virtual forces and torques.

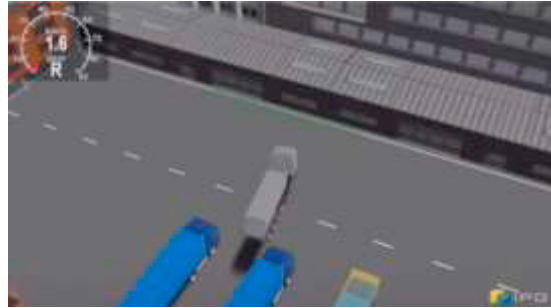


Fig. 2. Ford Otosan F-Max modelled in IPG simulation environment

The principle of d'Alembert is applied to get differential equations of motion for the generalized coordinates of the system, which can be written as:

$$A(q)\ddot{q} = B(q, \dot{q}) \quad (2)$$

where, A is the mass matrix containing the projection of various masses onto the free modes of motion and the vector $\ddot{q}$ contains the second-order derivatives of the generalized coordinates of the system. Additionally, to the generalized coordinates q_i , there exist further degrees of freedom, which are the steering rack position q_{steer} and the angular velocity w_{ij} of the wheel relative to the wheel carrier. The vehicle body encounters internal and external forces which are used in the calculation of the vehicle model. The vehicle body directly experiences gravitational and aerodynamic forces, gyroscopic, brake and transmission moments under kinematic and elasto-kinematic constraints due to the influence of the steering rack, (IPG TruckMaker Reference Manual V6.0.6 2017).

The co-simulation block diagram is shown in Fig. 3. In this diagram, the green and orange blocks are simulated using IPG, whereas the other ones are simulated with MATLAB/Simulink. The IPG blocks are interfaces to the build-in complex dynamic vehicle model. The orange interfaces read perception data from the surrounding IPG environment and vehicle and send signals to the vehicle (i.e., through a Simulink/IPG interface). Additional interfaces might need to be defined in MATLAB/Simulink between each block to adapt the data transferred between blocks.

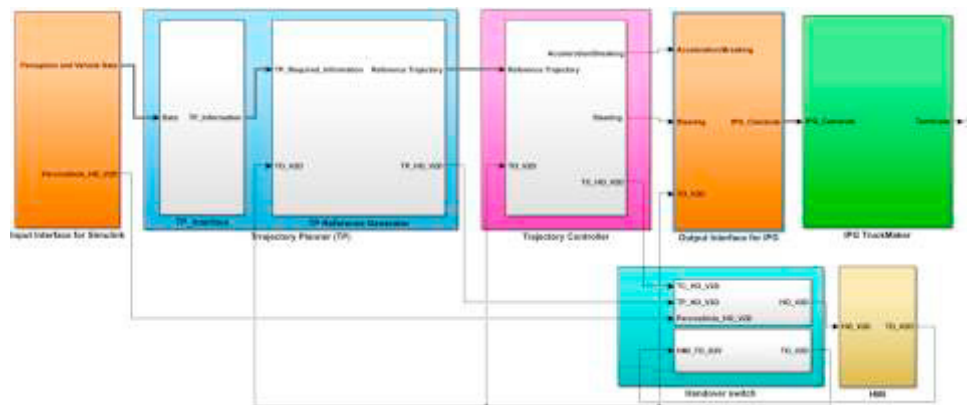


Fig. 3. Block diagram of the co-simulation platform based on IPG TruckMaker and Matlab/Simulink

The MATLAB/Simulink blocks in grey are used to implement and test different autonomous driving functions, such as trajectory planning and control. A trajectory planner, in blue, is used to generate an adequate reference trajectory considering the selected scenario. The planner might require an interface to pre-process the data of surroundings for the IPG environment and vehicle model at its inputs. The trajectory controller receives the planner's reference and provides the required longitudinal and lateral control signals for the vehicle to follow the reference. Finally, a handover switch block is dedicated to monitoring potential faults and malfunction of other blocks to initiate the control transition. Based on the situation the block switches the control from the driver to the Automated Driving System and vice-versa, allowing the use of this co-simulation architecture in various scenarios triggering different conditions.

2.3. Trajectory controller

This section presents the controllers used to overcome the unstable hitch joint problem of aHDVs is adopted from (C. Pradalier and K. Usher, 2008; M. Werling et al. 2014). The transfer functions for the hitch angle γ and trailer yaw ψ_{st} to the steering, input δ can be given as

$$G_{\gamma} = \frac{v_t \left(1 - \frac{L_{rh}}{L_{st}}\right)}{L_t \left(s + \frac{v_t}{L_{st}}\right)}, \quad G_{\psi_{st}} = \frac{v_t \left(\frac{L_{rh}}{L_{st}}s + \frac{v_t}{L_{st}}\right)}{L_t \left(s + \frac{v_t}{L_{st}}\right)} \quad (3)$$

Even though simple proportional gain can effortlessly stabilize the hitch angle (A. J. Rimmer and D. Cebon, 2016); A. J. Rimmer and D. Cebon, (2017); C. Pradalier and K. Usher, 2008), problem arises due to the zero of $G_{\psi_{st}}$. A comprehensive analysis of this issue is provided in (A. J. Rimmer and D. Cebon, 2016); however, the unstable zero injected from the transfer function $G_{\psi_{st}}$ could make the controller design non-trivial (S. Skogestad and I. Postlethwaite, 2005; K. J. Astrom, 2000). In (C. Pradalier and K. Usher, (2008)), a control for an agriculture truck trailer is proposed and implemented, inspired by the method developed in (P. Ridley and P. Corke, 2003). A scheme developed for trajectory tracking for forward motion is shown in Fig. 4 adapted and used in this paper. Unlike (P. Ridley and P. Corke, 2003), several assumptions are made as in (C. Pradalier and K. Usher, 2008).

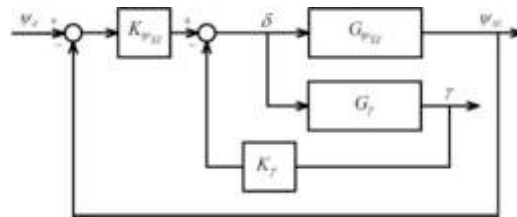


Fig. 4. Block diagram for a hitch stabilizing and trailer yaw tracking controller

3. Simulation results

Throughout the analysis an identical controller was implemented for both scenarios; the only difference was the use of steering ratio and low-level speed controller in the co-simulation model which are compensated in the hK model. The vehicle has a constant speed of -1m/s. In particular, the scenarios trend in the literature is to use a vehicle speed less than circa -1m/s (e.g., in expert driver tests, the speed is lower than -0.5m/s in a tight parking manoeuvre). Specifically, the longitudinal and steering dynamics are identified and reflected in the hK model using $\tau_v = 3$ s and $\tau_{\delta} = 0.18$ s. Fig. 5 shows the simulation results, in which step input is applied. The similarity between both models can be easily identified from the truck yaw, trailer yaw, hitch angle as well as steering angle. Apart from that, non-minimum phase behaviour resulting from the unstable zero is also evident.

In the second scenario, hK is evaluated by applying a sinusoidal reference equal to $10 \sin(0.1t)$. Test results are shown in Fig. 6. As in the first scenario, it is observed that both models have almost identical behavior.

It should be noted that the controller used can be enhanced further by integrating a feedforward controller, which is not the main focus of this paper.

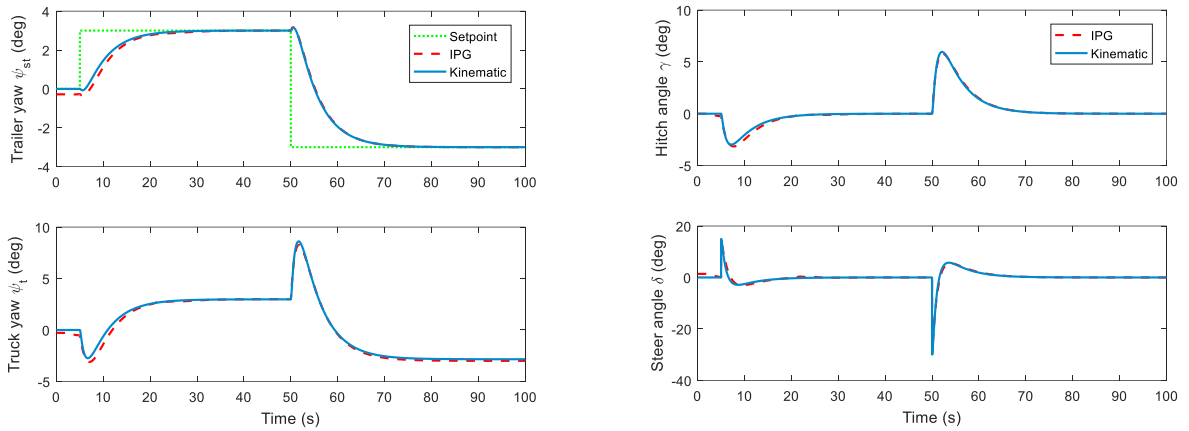


Fig. 5. Comparison of dynamic and hybrid kinematic model response for a stepwise reference in terms of a.) yaw of the trailer, b.) yaw of the truck, c.) hitch angle and d.) steering angle (green dashed line: reference set point of step function, red dashed line: dynamic vehicle model, blue line: hybrid kinematic model)

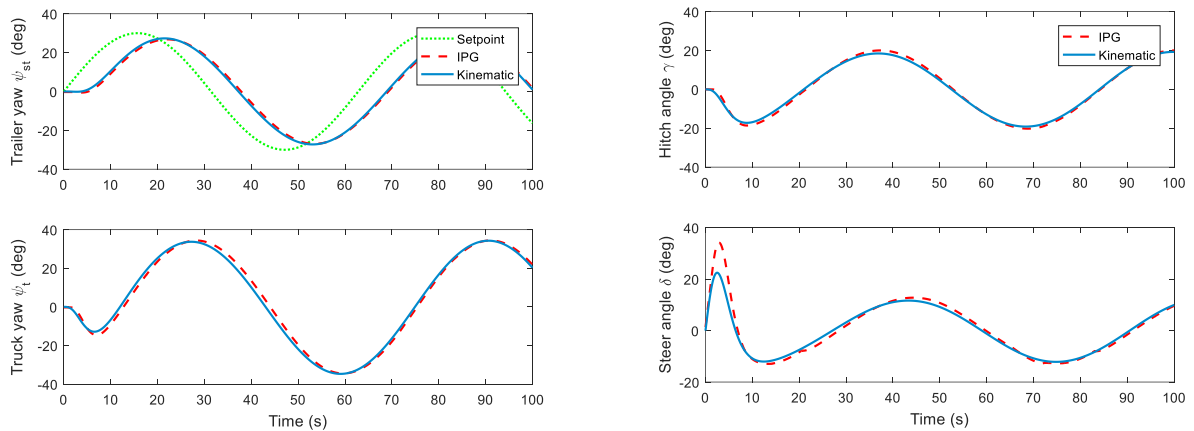


Fig. 6. Comparison between dynamic and hybrid kinematic models for a sinusoidal input in terms of a.) yaw of the trailer, b.) yaw of the truck, c.) hitch angle and d.) steering angle (green dashed line: reference set point of step function, red dashed line: dynamic vehicle model, blue line: hybrid kinematic model)

4. Conclusion and future works

To conclude, the challenges associated with articulated HDVs lead to complex dynamics and therefore require appropriate models to fully capture the vehicle's response. Throughout the paper, practical points and implicit consensus are highlighted. Research in the literature agrees on choosing kinematic models for the design of autonomous functions for reversing autonomous HDVs. The simulation results of this paper showed the validity of this choice. Simulation results show that the hK model captures the actual vehicle behavior that is required to design a tracking controller for an autonomous articulated vehicle for challenging scenarios such as reverse parking as well as is advantageous nature due to the incorporation of transfer functions, delays, and saturations to mimic actual vehicle behavior. The hK model will be used in future work to develop algorithms for the automation of heavy-duty vehicles in more complex scenarios, such as minimum risk maneuver.

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